

ChronoShore: Diegetic Exploration of Short- and Long-Term Environmental Time in a Virtual Coastal Ecosystem

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Figure 1: Overview of the ChronoShore environment, showing the simulated shoreline with interactable palm trees used as environmental timekeepers.

ABSTRACT

We present an extended version of ChronoShore, an immersive virtual reality (VR) environment for interacting with environmental time at multiple temporal scales. The virtual environment is based on a multi-resolution reconstruction of a real segment of the Caribbean coast of Costa Rica, generated from heterogeneous geospatial datasets including aerial point clouds, satellite-based elevation data, and global bathymetry. Unlike conventional timeline-based controls, ChronoShore enables users to explore and manipulate time through diegetic environmental cues, or *zeitgebers*, embedded directly in the virtual world.

In addition to short-term time control via celestial bodies such as the sun and moon, the new version introduces long-term temporal interaction through vegetation, particularly trees, allowing users to traverse years or decades within the simulation. The environment features an enhanced weather model with realistic seasonal dynamics, including rain patterns and cloud morphologies that evolve over the course of the day in response to atmospheric conditions.

ChronoShore serves as a research and application platform for studying time perception in VR, immersive storytelling of geological and environmental processes, and relaxation-oriented experiences. The platform is available to other researchers upon request.

1 INTRODUCTION

Understanding and experiencing temporal processes is a central challenge in immersive simulations, particularly when these processes unfold across vastly different time scales. In many interactive systems, time

navigation is achieved through abstract interface elements such as sliders or scrub bars. While effective in desktop-based applications, these metaphors translate poorly to virtual reality, where they break immersion and provide little contextual grounding for the passage of time.

ChronoShore explores an alternative approach by embedding time interaction directly into the virtual environment. Instead of abstract controls, users manipulate time through environmental elements that naturally convey temporal information, commonly referred to as *zeitgebers* [8]. In earlier work, we demonstrated how celestial bodies could be used to control time progression over the span of days, enabling users to observe changes in weather, lighting, and atmosphere within a simulated coastal scene [5].

This paper presents a significantly expanded version of ChronoShore (see Figure 1; <https://youtu.be/3cBksmfy2jo>). The simulated coastal landscape is not manually modeled but reconstructed from real geospatial data. It represents a critical segment of the Caribbean coast of Costa Rica, an area affected by ongoing shoreline change and erosion. A multi-resolution terrain model combines high-resolution aerial measurements in the coastal zone with lower-resolution inland elevation and offshore bathymetric data, enabling both local detail for first-person exploration and large-scale geographic context.

Weather simulation has been refined to reflect seasonal dynamics, particularly those characteristic of tropical rainy seasons, with cloud formations that change their morphology over the course of the day based on temperature and atmospheric conditions.

Most notably, ChronoShore now supports interaction with environmental elements representing long-term temporal processes. Vegetation, and especially trees, functions as a new class of interactable *zeitgebers*, allowing users to move forward and backward across years. This enables seamless transitions between short-term and long-term temporal exploration within the same immersive experience.

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Figure 2: Reconstructed virtual landscape of the Caribbean coast of Costa Rica, generated from geological and topographical data.

2 ENVIRONMENT DESIGN AND SIMULATION

A primary design goal of the extended ChronoShore environment was to achieve a high degree of realism while maintaining real-time performance in VR. The environment (see Figure 2) is constructed using a reproducible geospatial processing pipeline (see [6] for more details) that integrates datasets of varying resolution and spatial extent into a continuous terrain representation. High-resolution aerial data is used in perceptually critical coastal regions, while satellite-based elevation and global bathymetry provide contextual coverage inland and offshore. This hierarchical representation supports real-time performance while preserving perceptual continuity at human scale.

The ocean simulation builds upon FFT-based wave generation using a modified JONSWAP spectrum [4]. Multiple wave cascades at different spatial scales are combined to reduce tiling artifacts and to support continuous time manipulation, including fast-forwarding and rewinding. Wave behavior dynamically adapts to changing weather conditions, such as increased wind strength during storms.

Atmospheric rendering is based on a physically-based sky model [2], enabling realistic changes in lighting and sky color throughout the day. Volumetric clouds, inspired by techniques used in modern games [9, 7], evolve over time in response to simulated temperature and humidity, resulting in distinct cloud morphologies during calm and rainy periods.

Vegetation placement is informed by terrain-derived masks encoding elevation, slope, and surface characteristics. Trees and other plants are authored with lightweight vertex animations to represent wind interaction and growth-related changes over longer time spans. Across the environment, visual properties such as material wetness, ambient lighting, and color saturation respond continuously to weather and time, contributing to a cohesive and immersive experience.

3 TIME INTERACTION AND ENVIRONMENTAL ZEITGEBERS

ChronoShore is designed around the principle that time interaction should be grounded in elements that users intuitively associate with temporal change. Short-term time manipulation is achieved through interaction with celestial bodies, primarily the sun and moon, whose motion is computed using analytical models of planetary movement [1] and Earth rotation [3]. By grabbing and moving these bodies along their trajectories, users can scrub through hours or days while remaining spatially situated within the environment.

To support longer temporal jumps, the extended system introduces vegetation-based zeitgebers. Trees encode long-term temporal information through changes in size, shape, and appearance, allowing users to traverse years or decades by interacting with them. Because vegetation is embedded within a geographically grounded landscape, interactions with trees connect long-term temporal change to a spatial context that reflects real coastal morphology rather than an abstract scene.

This dual-scale approach enables smooth transitions between short-term phenomena, such as storms, and long-term environmental

and geological change within a single coherent interface.

Because some zeitgebers may be occluded or temporarily absent from view, optional visualizations of celestial trajectories and growth timelines can be enabled. These visual aids preserve the diegetic nature of the interaction while ensuring accessibility and usability in complex weather conditions.

4 APPLICATIONS AND RESEARCH DIRECTIONS

ChronoShore is currently used as a platform for multiple research and application domains. One focus is the investigation of time perception in immersive virtual environments, particularly how different types of zeitgebers influence users' understanding of temporal scale and duration.

Beyond interaction design and research in perception, ChronoShore serves as a platform for environmental communication and digital heritage exploration. By embedding time manipulation within a reconstructed coastal landscape, the system enables experiential storytelling of geological and environmental processes such as shoreline change, seasonal weather dynamics, and long-term ecological transformation. This makes the platform suitable for research, education, and public outreach contexts concerned with environmental awareness.

Finally, informal observations from earlier demonstrations [5] indicated that users often perceive interaction with the system as calming and meditative. The seamless, continuous manipulation of time through natural environmental elements appears to promote relaxation, motivating ongoing exploration of ChronoShore as a relaxation-oriented VR experience.

The ChronoShore platform is available to other researchers interested in studying time, environmental simulation, or immersive interaction. Interested parties are encouraged to contact the authors for access.

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